TrES-1b

R-1176

Benchmark run · workflow W8-benchmark-deep · record `deep-bench_TRES-1_211101` · created 2026-07-31T03:49:26+00:00

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459519.6574 +/- 0.0018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.016
Transit depth (Rp/Rs)^2	2.24 +/- 0.49 [%]
Orbital Inclination (inc)	85.95 +/- 1.25
Airmass coefficient 1 (a1)	0.973 +/- 0.01
Airmass coefficient 2 (a2)	0.0144 +/- 0.002
Scatter in the residuals of the lightcurve fit is	1.08 %
Best Comparison Star	#2 - [512, 128]
Optimal Aperture	4.43
Optimal Annulus	6.44
Transit Duration (day)	0.075 +/- 0.024

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.016 vs 0.1358 (deviation 0.82σ)
Duration fit vs published	0.075 d vs 0.1042 d (deviation 1.12σ)
Mid-time O–C	6.2 min
Depth SNR	8.6
Residual scatter	1.08 %

Light curve

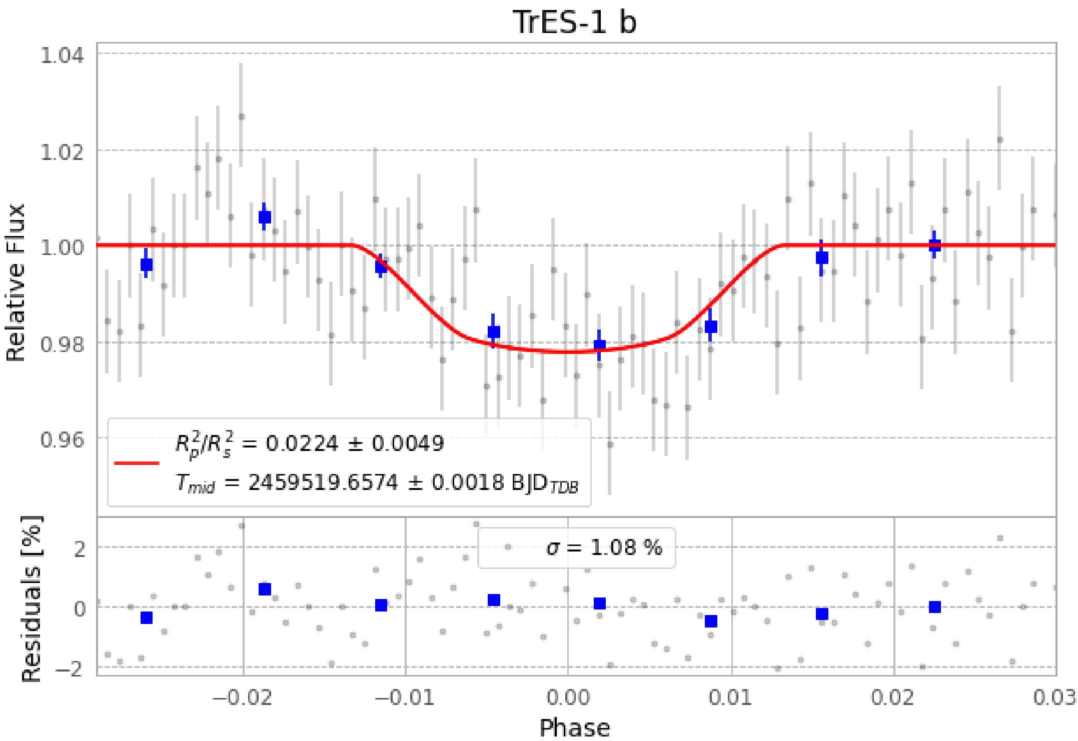

Figure 1 — FinalLightCurve_TrES-1 b_2021-10-31.png (SHA-256 1f2c7ca595b0e461…)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [423, 261], 3 comparison star(s). Exact configuration: `parameters.inits_json` in record.json (SHA-256 31ac97fd6b511478…).

`exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b`

Data provenance

87 FITS frames (57 MB), `TRES-1211101013912.FITS` → `TRES-1211101055712.FITS`. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-1-211101.log (11a2a8cb1aa3…), FinalLightCurve_TrES-1 b_2021-10-31.png (1f2c7ca595b0…), FinalLightCurve_TrES-1 b_2021-10-31.csv (486be551e401…)

Compute resources

Wall time 145.3 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)